<table>
<tr><th colspan="3">C2.2 What does the Periodic Table tell us about the elements?</th></tr>
<tr><th>Teaching and learning narrative</th><th colspan="2">Assessable learning outcomes
Learners will be required to:</th></tr>
<tr><td rowspan="9">Elements in the modern Periodic Table are arranged in periods and groups, based on their atomic numbers. Elements in the same group have the same number of electrons in their outer shells. The number of electron shells increases down a group but stays the same across a period.

Mendeleev proposed the first arrangement of elements in the Periodic Table. Although he did not know about atomic structure, he reversed the order of some elements with respect to their masses, left gaps for undiscovered elements and predicted their properties. His ideas were accepted because when certain elements were discovered they fitted his gaps and the development of a model for atomic structure supported his arrangement. The later determination of the number of protons in atoms provided an explanation for the order he proposed (IaS3).

The Periodic Table shows repeating patterns in the properties of the elements. Metals and non-metals can be identified by their position in the Periodic Table and by comparing their properties (physical properties including electrical conductivity).

Properties of elements within a group show trends. The reactivity of Group 1 metal elements increases down the group, shown by their reactivity with moist air, water and chlorine.

The Group 7 halogens are non-metals and become less reactive down the group. This is shown in reactions such as their displacement reactions with compounds of other halogens in the group.</td><td>1.</td><td>explain how the position of an element in the Periodic Table is related to the arrangement of electrons in its atoms and hence to its atomic number</td></tr>
<tr><td>2.</td><td>describe how Mendeleev organised the elements based on their properties and relative atomic masses</td></tr>
<tr><td>3.</td><td>describe how discovery of new elements and the ordering elements by atomic number supports Mendeleev's decisions to leave gaps and reorder some elements</td></tr>
<tr><td>4.</td><td>describe metals and non-metals and explain the differences between them on the basis of their characteristic physical and chemical properties, including melting point, boiling point, state and appearance, density, formulae of compounds, relative reactivity and electrical conductivity</td></tr>
<tr><td>5.</td><td>recall the simple properties of Group 1 elements including their reaction with moist air, water, and chlorine</td></tr>
<tr><td>6.</td><td>recall the simple properties of Group 7 elements including their states and colours at room temperature and pressure, their colours as gases, their reactions with Group 1 elements and their displacement reactions with other metal halides</td></tr>
<tr><td>7.</td><td>predict possible reactions and probable reactivity of elements from their positions in the Periodic Table</td></tr>
<tr><td>8.</td><td>describe experiments to identify the reactivity pattern of Group 7 elements including displacement reactions
PAG1</td></tr>
<tr><td>9.</td><td>describe experiments to identify the reactivity pattern of Group 1 elements</td></tr>
</table>

Linked learning opportunities

Practical work:

- reactions of Group 1 (demonstration) and Group 7 (for example displacement)

Ideas about Science:

- understanding how scientific explanations and models develop, in the context of the Periodic Table (IaS3)

Ideas about Science:

- making and testing predictions about trends and patterns in the Periodic Table (IaS1)